

Academic Elective Bidding & Allocation System

Problem Statement #05 — Campus & Academic Operations

Lab 1: Requirements Engineering & UML Use-Case Modelling

Target Stakeholders / Actors: Student • Academic Registrar • Notification Service (secondary)

University departments need an automated elective course selection mechanism where students allocate bidding credits across preferences and the engine solves allocation constraints while preventing timetable collisions.

Deliverables:

1. Complete Requirements Table (5 FRs + 2 NFRs)
2. UML Use-Case Diagram
3. Use-Case Flow Specification

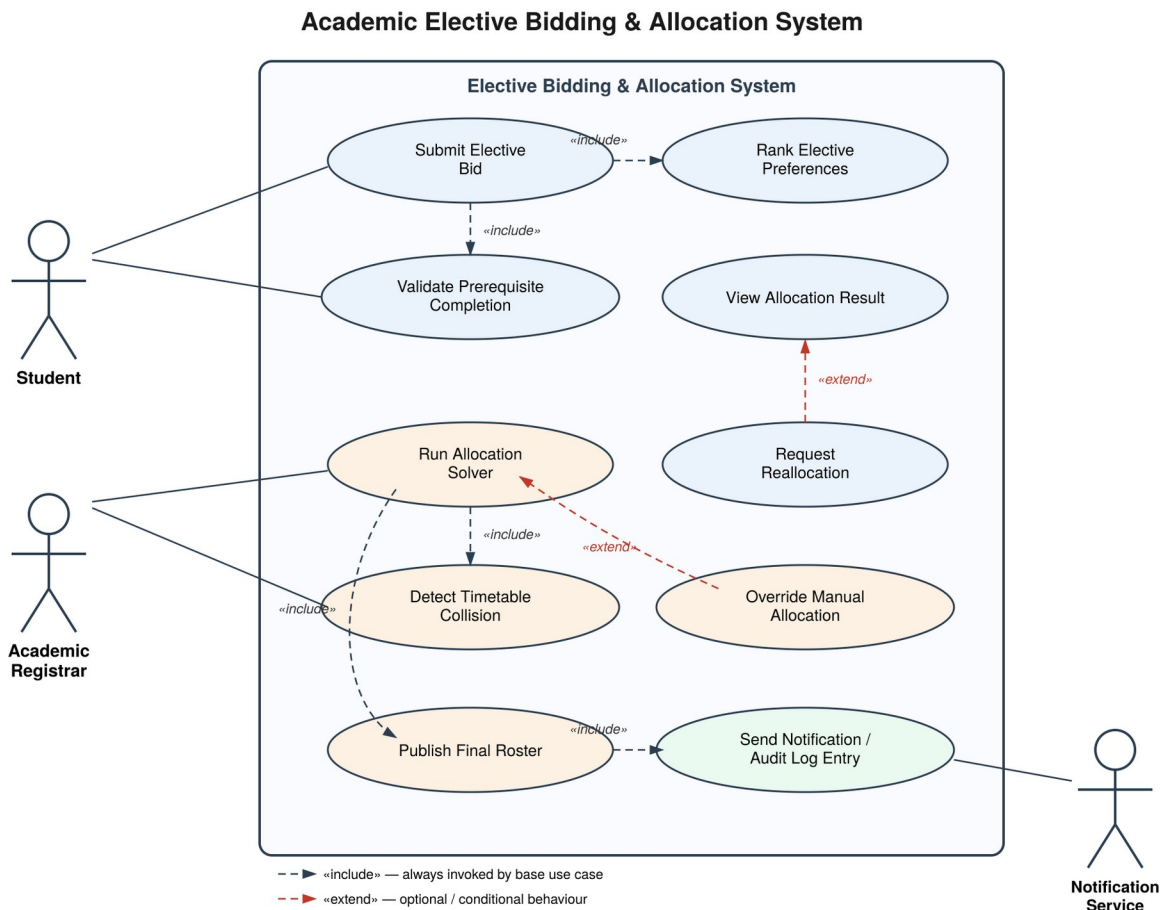
1. Requirements Table

Five Functional Requirements (FR-001 to FR-005) and two Non-Functional Requirements (NFR-001, NFR-002). FR-001 and NFR-001 are the instructor-provided baseline requirements; all other rows are newly authored for this system.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow students to distribute 100 bidding credits across their ranked elective preferences and validate prerequisite course completions.	High	Pass: Credits deduct accurately and prerequisites validated. Fail: Total allocated credits exceed 100 or unmet prerequisites bypassed.	Core bidding mechanism; guarantees valid, comparable input for the allocation solver.
FR-002	Functional	The system shall allow students to rank all selected electives in strict preference order and assign a credit value to each ranked choice before final submission.	High	Pass: Every selected elective has a unique rank and a credit value at submission time. Fail: A bid is accepted with an unranked or unweighted elective.	Provides the solver with complete, ordered preference data needed to compute optimal allocation.
FR-003	Functional	The system shall detect and block submission of any elective bid whose class timing overlaps with the student's existing fixed/core-course timetable.	High	Pass: System identifies the conflicting slot and prevents submission when an overlap exists. Fail: A bid with an overlapping timetable slot is accepted.	Prevents students from being allocated electives that are physically impossible to attend.
FR-004	Functional	The system shall place students who are not allocated a seat on a waitlist and automatically reallocate any seat vacated by a drop or swap to the next eligible waitlisted student.	Medium	Pass: A freed seat is auto-assigned to the highest-priority waitlisted student within one reallocation cycle, with notification sent within 5 minutes. Fail: A vacated seat remains unfilled after a completed reallocation cycle.	Maximizes seat utilization and gives waitlisted students a fair, transparent path to allocation.
FR-005	Functional	The system shall allow the Academic Registrar to manually override a solver-generated allocation for a specific student, requiring a mandatory justification note stored against that decision.	Medium	Pass: Override is applied, reflected in the student's final roster, and the justification note is stored and retrievable. Fail: An override is applied with no justification note recorded.	Supports legitimate exceptions (e.g., accommodations, administrative corrections) while preserving auditability.
NFR-001	Nonfunctional	The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures the allocation engine remains usable at university scale without delaying result publication.
NFR-002	Nonfunctional	The system shall maintain an immutable, timestamped audit log of every bid submission, allocation change, and registrar override, retained for a minimum of three academic years.	High	Pass: Every state-changing action produces a tamper-evident log entry retrievable in under 2 seconds. Fail: Any allocation-affecting action occurs without a corresponding log entry.	Required for academic-integrity dispute resolution, grievance handling, and institutional compliance audits.

2. UML Use-Case Diagram

Actors: Student, Academic Registrar (primary); Notification Service (secondary system actor). The diagram models the primary use cases of the bidding and allocation workflow, with «include» relationships for mandatory sub-behaviour and «extend» relationships for optional/conditional behaviour.



Relationship Summary

- «include» — Submit Elective Bid always invokes Rank Elective Preferences and Validate Prerequisite Completion.
- «include» — Run Allocation Solver always invokes Detect Timetable Collision and (on completion) Publish Final Roster.
- «include» — Publish Final Roster always invokes Send Notification / Audit Log Entry.
- «extend» — Request Reallocation optionally extends View Allocation Result (only if the student is unsatisfied/unallocated).
- «extend» — Override Manual Allocation optionally extends Run Allocation Solver (only invoked by the Registrar for exception handling).

3. Use-Case Flow Specification

Use Case: **Submit Elective Bid**

Primary Actor: Student

Includes: Rank Elective Preferences, Validate Prerequisite Completion

Preconditions

- The student is authenticated and currently enrolled for the academic term.
- The elective bidding window is open, as per the published academic calendar.
- The student's core-course (fixed) timetable has already been finalized in the system.

Postconditions

- Success: The student's ranked preferences and credit distribution are recorded, locked, and time-stamped; total credits allocated equal exactly 100; an audit log entry is created.
- Failure: No bid is recorded; the student retains unused bidding credits and may resubmit before the window closes.

Main Success Scenario

1. Student selects "Submit Elective Bid" from the dashboard.
2. System displays available electives with seat availability and prerequisite status.
3. Student ranks the desired electives in strict preference order (includes Rank Elective Preferences).
4. Student distributes 100 bidding credits across the ranked preferences.
5. System validates prerequisite completion for each selected elective against the student's transcript (includes Validate Prerequisite Completion).
6. System checks the selected electives for collisions against the student's fixed timetable.
7. System verifies that total allocated credits equal exactly 100.
8. Student reviews the bid summary and confirms submission.
9. System time-stamps and locks the bid, and generates an audit log entry.
10. System displays a confirmation screen with a unique bid receipt reference number.
11. Use case ends successfully.

Alternate Flow — A1: Credit Total Invalid or Timetable Collision Detected (at step 6–7)

- A1.1 If a selected elective's timing overlaps the student's fixed timetable, the system flags the specific conflicting elective and blocks submission.
- A1.2 If total allocated credits do not equal 100, the system highlights the discrepancy and blocks submission.
- A1.3 Student edits preferences and/or credit distribution to resolve the flagged issue(s).
- A1.4 Flow resumes at step 6.

Alternate Flow — A2: Prerequisite Not Met (at step 5)

- A2.1 System marks the elective as ineligible and excludes it from ranking.
- A2.2 Student is prompted to choose a replacement elective or proceed with fewer ranked choices.
- A2.3 Flow resumes at step 3.